

There Are No Collatz m -Cycles with $m \leq 96$

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Abstract

A nontrivial periodic orbit of the shortcut Collatz map is called an m -cycle when one period contains exactly m local minima. We prove that no nontrivial positive Collatz m -cycle exists for $92 \leq m \leq 96$. Together with Hercher's exclusion for $m \leq 91$, this implies that every nontrivial positive Collatz cycle would have at least 97 local minima.

The proof combines a Diophantine reduction with a finite exact computation. Barina's verification bound, Hercher's growth and odd-step estimates, and the Simons–de Weger upper bound first imply, through a suffix-balanced block estimate and an exact Farey-neighbor argument, that every hypothetical cycle in the required range satisfies

$$K \geq 205632218873398596256,$$

where K is the total number of odd shortcut steps. For each $m = 92, \dots, 96$, a finite interval for the least local minimum is then exhausted by an affine-ladder search. The search consists of 372 disjoint root branches and 464 final work units; independent C++/GMP and Rust/rug implementations agree on all counters and find no survivor. Above the finite intervals, a second denominator lift yields a lower bound for K that contradicts the Simons–de Weger upper bound. The cases $m = 92, \dots, 95$ were obtained independently but overlap Wang's 2026 preprint, which the author found during a later literature review; no priority claim is made here for those four exclusions. The new cycle-exclusion case is $m = 96$.

Keywords: Collatz problem, m -cycles, continued fractions, Diophantine approximation, computer-assisted proof, exact certificate

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1. Introduction

The Collatz problem asks whether every positive integer reaches the cycle $1, 2$ under the shortcut map

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}. \end{cases} \quad (1)$$

The conjecture remains open. We consider only the periodic-orbit subproblem. A nontrivial positive periodic orbit is naturally decomposed into alternating runs of odd and even shortcut steps. If a period contains m local minima, the orbit is called an m -cycle.

Simons and de Weger developed strong Diophantine and computational restrictions on hypothetical m -cycles [5, 6]. Hercher subsequently proved that no nontrivial positive m -cycle exists for $m \leq 91$ [2, 3]. The argument and computations reported here for $m = 92, \dots, 96$ were developed independently. During a later literature review, after completion of the five-case computation, the author became aware of Wang's 2026 preprint excluding through $m = 95$ [7]. The present paper therefore makes no priority claim for the overlapping exclusions $m = 92, \dots, 95$; it proves the next case $m = 96$ and supplies one uniform exact certificate for all five cases.

Theorem 1.1 (Main theorem). *There is no nontrivial positive Collatz m -cycle for any*

$$92 \leq m \leq 96.$$

Corollary 1.2 (Record through $m = 96$). *There is no nontrivial positive Collatz m -cycle with $m \leq 96$. Equivalently, every nontrivial positive Collatz cycle has at least 97 local minima.*

Proof. Hercher excludes $m \leq 91$ [2, 3]; apply Theorem 1.1 to the five remaining cases. $\square$

1.1. Proof strategy

The argument is a two-sided reduction around a finite search. First, a cyclic suffix-balancing lemma and Hercher's growth estimate convert a lower bound for the total odd-step count K into lower bounds for a long block of local minima. Inserting these bounds into the cycle product inequality gives

a short interval immediately to the right of $\log_2 3$. Exact Farey adjacency then forces the common lower bound

$$K \geq K_0 := 205632218873398596256. \quad (2)$$

For each $m \in \{92, 93, 94, 95, 96\}$ we then split according to the least local minimum n_1 . If n_1 lies in the finite interval listed in Table 1, an exact affine-ladder search excludes it. If n_1 lies above that interval, every local minimum is larger, and the same denominator argument gives

$$K \geq Q := 7941964418702608664581, \quad (3)$$

whereas the Simons–de Weger estimate gives $K < Q$.

The finite computation is used only after completeness of the search decomposition has been proved. The valuation split, one-bit Hensel split, interval cuts, deterministic tail, and subdivision into work units are exact partitions of the reduced search space. The released C++ and Rust engines are independent implementations of this mathematical specification. Their agreement reduces implementation risk; it does not replace the completeness proof.

1.2. Independent development and relation to Wang’s preprint

The five-case argument and computations reported here were developed independently, before the author became aware of Wang’s preprint [7]. The overlap was identified only during a subsequent literature review undertaken while revising the manuscript. This chronology is stated to distinguish independent development from methodological dependence; it is not a priority claim. Wang’s preprint provides an earlier public record for the exclusions $m = 92, \dots, 95$, and no priority is claimed here for those four exclusion results.

A retrospective comparison shows substantial structural convergence. For $m = 92, \dots, 95$, both works use a suffix-balanced cyclic block estimate, a denominator-lift mechanism based on rational approximation to $\log_2 3$, and an affine a_1 -ladder for the finite search. The finite windows and search depths for those four cases also agree, and both treatments reach the thresholds K_0 and Q , which are dictated by adjacent rational approximants to $\log_2 3$. These agreements were discovered after the present computation had been completed.

The present artifact retains the four overlapping cases because they form part of the independently completed five-case computation and provide a

uniform independent certification benchmark with two exact engines. The analytic route also differs in two respects relevant to that uniform certificate: Hercher’s stronger lower bound is used to reach K_0 in one denominator lift, and the later stage minima are obtained from the remaining odd-step mass after the preceding valuation caps have been removed, while the cycle remains rotated at its global minimum. The new cycle-exclusion case proved here is $m = 96$. In that case the finite reduction uses the window $[X, 29X]$, depth 7, 75 root branches, and 113 final work units; these figures arise from the same certified reduction rather than from a separate method introduced after the overlapping cases.

1.3. Published inputs and computational role

The proof uses the following published inputs.

- (I1) Barina’s exhaustive verification implies that every member of a nontrivial positive cycle is at least

$$X = 2^{71} = 2361183241434822606848$$

[1].

- (I2) Hercher’s local-minimum growth estimate is used in Lemma 2.3, and the odd-step lower bound gives $K > 7.76 \times 10^{19}$ in the range needed here [2, 3]. All citations to Hercher’s 2023 paper are to be read together with the 2026 corrigendum.

- (I3) In the required range, Simons and de Weger give

$$K < 1.4784 m (\log_2 3)^m$$

[5, 6].

The manuscript proves the suffix-balanced reduction, the cap-complement stage bounds, and the exact search-partition lemma. The accompanying artifact [9] checks all derived numerical inequalities with exact integer or rational arithmetic, including logarithm enclosures, Farey determinants, finite windows, valuation caps, stage minima, prefix margins, branch and work-unit coverage, and the final upper-bound comparisons. It also authenticates source and binary hashes, build and execution provenance, matching dual-engine

counters, zero hits, and zero unresolved units. No floating-point result is used in a mathematical decision.

The remainder of the paper is organized as follows. Section 2 records the block model and the published inputs. Sections 3 and 4 give the common denominator lift to K_0 . Section 5 derives the finite windows, caps, and rotation-correct stage minima. Section 6 proves the short-prefix pruning rule. Section 7 describes the affine-ladder search and proves completeness. Section 8 records the certified computation, and Section 9 gives the final contradiction.

2. The block model and published inputs

Let $n_1, \dots, n_m$ be the local minima of a hypothetical nontrivial cycle, indexed cyclically. Starting at n_i , let $k_i \geq 1$ be the number of consecutive odd shortcut steps and let $\ell_i \geq 1$ be the number of following even shortcut steps before n_{i+1} . Put

$$K = \sum_{i=1}^m k_i, \quad L = \sum_{i=1}^m \ell_i, \quad \delta = \log_2 3. \quad (4)$$

The number K is also the odd-state period of the associated accelerated orbit.

Lemma 2.1 (Local-minimum transition). *For every block there is an odd positive integer a_i such that*

$$n_i = 2^{k_i} a_i - 1 \quad (5)$$

and

$$n_{i+1} = \frac{3^{k_i} a_i - 1}{2^{\ell_i}}, \quad \ell_i = \nu_2(3^{k_i} a_i - 1), \quad k_{i+1} = \nu_2(n_{i+1} + 1). \quad (6)$$

Proof. The exact odd-run length k_i means that $2^{k_i} \mid n_i + 1$ and that $(n_i + 1)/2^{k_i}$ is odd; call it a_i . Repeated use of the odd branch of (1) gives $C^{k_i}(n_i) = 3^{k_i} a_i - 1$. Dividing by the exact following power 2^{ℓ_i} gives the next odd local minimum. The formula for k_{i+1} follows from (5) at the next block. $\square$

For a local minimum n whose odd-run length is k , define

$$T(n) = \sum_{j=0}^{k-1} \frac{1}{C^j(n)}.$$

Lemma 2.2 (Cycle product inequality). *Every nontrivial positive cycle satisfies*

$$0 < \frac{K+L}{K} - \delta < \frac{1}{3K \log 2} \sum_{i=1}^m T(n_i), \quad T(n_i) < \frac{3}{n_i}. \quad (7)$$

Proof. Multiplication of the step ratios around the cycle gives

$$1 = \frac{3^K}{2^{K+L}} \prod_{\text{odd states } x} \left(1 + \frac{1}{3x}\right).$$

Taking natural logarithms yields

$$(K+L) \log 2 - K \log 3 = \sum_{\text{odd states } x} \log \left(1 + \frac{1}{3x}\right) > 0.$$

Using $\log(1+u) < u$ and grouping the odd states by blocks proves the first estimate. During an odd run, $C(x) > (3/2)x$; hence

$$T(n) < \frac{1}{n} \sum_{j \geq 0} \left(\frac{2}{3}\right)^j = \frac{3}{n}.$$

□

We use the following consequence of Hercher's growth lemma.

Lemma 2.3 (Block growth). *For consecutive local minima $n_j, n_{j+1}, \dots, n_{j+s-1}$,*

$$\sum_{r=0}^{s-1} k_{j+r} < \log_2(n_j + 1) \frac{\delta^s - 1}{\delta - 1}. \quad (8)$$

Consequently, if the sum on the left is at least H , then

$$\log_2(n_j + 1) > H \frac{\delta - 1}{\delta^s - 1}. \quad (9)$$

Proof. Hercher's Lemma 20 gives $n_{i+1} < n_i^\delta$. Since $\delta > 1$,

$$n_{i+1} + 1 < n_i^\delta + 1 < (n_i + 1)^\delta.$$

Together with $2^{k_i} \mid n_i + 1$, this gives

$$k_{j+r} \leq \log_2(n_{j+r} + 1) < \delta^r \log_2(n_j + 1)$$

for $r \geq 1$, while $k_j \leq \log_2(n_j + 1)$. Summing the geometric progression proves (8); rearrangement gives (9). □

The numerical inputs used throughout are

$$X = 2^{71} = 2361183241434822606848, \quad K \geq K_* := 77600000000000000000 \quad (10)$$

for $92 \leq m \leq 96$, and

$$K < 1.4784 m \delta^m. \quad (11)$$

The strict published lower bound $K > 7.76 \times 10^{19}$ safely implies the integer inequality in (10). For exact upper comparisons we use

$$\delta < \bar{\delta} := \frac{317}{200}, \quad 1.4784 = \frac{924}{625}. \quad (12)$$

3. A suffix-balanced denominator estimate

Lemma 3.1 (Suffix-balanced cyclic rotation). *Let $k_1, \dots, k_m$ be positive real numbers with sum K . After a cyclic rotation,*

$$\sum_{j=m-t+1}^m k_j \geq \frac{tK}{m} \quad (1 \leq t \leq m). \quad (13)$$

Proof. Set $d_i = k_i - K/m$, so that $\sum_i d_i = 0$. Choose a cyclic origin immediately after a maximum of the cyclic partial sums of the d_i . Every prefix sum in the new order is nonpositive. Since the total sum is zero, every suffix sum is nonnegative, which is (13). $\square$

The lemma is elementary. Its use in the present m -cycle reduction is shared with Wang [7]. Fix the rotation of Lemma 3.1. Applying Lemma 2.3 to the suffix of length t gives

$$\log_2(n_{m-t+1} + 1) > \frac{tK}{m} \frac{\delta - 1}{\delta^t - 1}. \quad (14)$$

For $t \geq 1$, the function

$$h_t(x) = \frac{x - 1}{x^t - 1}$$

is decreasing on $x > 1$ (with $h_1 \equiv 1$). Hence $\delta < \bar{\delta}$ may be used conservatively in the right-hand side.

Definition 3.2 (Certified complement bound). Let $K^* > 0$, let Y_0 be a strict lower bound for all local minima, and choose $0 \leq s \leq m$. For $1 \leq t \leq s$, let f_t be the greatest integer strictly below

$$\frac{tK^* \bar{\delta} - 1}{m \bar{\delta}^t - 1}. \quad (15)$$

Let λ be a certified rational lower bound for $\log 2$. Define

$$E_m(K^*, Y_0, s) = \frac{1}{3K^*\lambda} \left(\frac{3(m-s)}{Y_0} + \sum_{t=1}^s \frac{3}{2^{f_t} - 1} \right). \quad (16)$$

Proposition 3.3 (Denominator-lift interval). *If a hypothetical m -cycle has $K \geq K^*$ and every local minimum is at least Y_0 , then*

$$0 < \frac{K + L}{K} - \delta < E_m(K^*, Y_0, s). \quad (17)$$

Proof. Choose a suffix-balanced block of length s . For its element whose terminal suffix has length t , (14), $K \geq K^*$, and monotonicity of h_t imply

$$\log_2(n+1) > \frac{tK^*}{m} h_t(\bar{\delta}) > f_t.$$

Thus $n > 2^{f_t} - 1$, and Lemma 2.2 bounds its contribution by $3/(2^{f_t} - 1)$. The remaining $m - s$ minima contribute less than $3(m - s)/Y_0$. Finally, $K \geq K^*$ and $\lambda < \log 2$ make (16) a conservative upper bound in (7). $\square$

Lemma 3.4 (Farey denominator bound). *Let $a/b < c/d$ be reduced fractions with $bc - ad = 1$. Every reduced fraction strictly between them has denominator at least $b + d$.*

Proof. If $a/b < p/q < c/d$, then $pb - aq \geq 1$ and $cq - pd \geq 1$. Hence

$$q = q(bc - ad) = b(cq - pd) + d(pb - aq) \geq b + d.$$

$\square$

4. The common first denominator lift

Theorem 4.1 (First denominator lift). *Every hypothetical positive m -cycle with $92 \leq m \leq 96$ satisfies*

$$K \geq K_0 = 205632218873398596256.$$

Proof. Take $K^* = K_*$, $Y_0 = X$, and $s = 88$. For $92 \leq m \leq 96$, the expression in (16) is largest at $m = 96$: the number $m - s$ of unselected minima increases with m , while the exponents in (15) decrease as m increases. It therefore suffices to certify the case $m = 96$.

The exact certificate encloses δ and $\log 2$ by outward rational intervals and proves that

$$\left(\delta, \delta + E_{96}(K_*, X, 88)\right)$$

lies strictly between the adjacent reduced fractions

$$\frac{202780263237295321099}{127940101513462006853} < \frac{123139092617126647266}{77692117359936589403}. \quad (18)$$

Their cross-determinant is 1, and their denominator sum is

$$127940101513462006853 + 77692117359936589403 = 205632218873398596256.$$

The reduced denominator of $(K + L)/K$ divides K . Apply Lemma 3.4 and Proposition 3.3. $\square$

5. Finite reductions: windows, caps, and stage minima

Rotate a hypothetical cycle so that n_1 is its least local minimum. We use the block coordinates of Lemma 2.1. For each case, choose a finite window

$$X \leq n_1 \leq W_m X.$$

The values of W_m are chosen so that the second denominator lift of Section 9 becomes contradictory as soon as the finite window is excluded.

Table 1: Certified finite reductions. The root branches are the possible values of $k_1 = \nu_2(n_1 + 1)$.

m	finite window	branches	depth	encoded caps $(\kappa_1, \dots, \kappa_d)$
92	$[X, \frac{73}{10}X]$	73	2	(73, 118)
93	$[X, 15X]$	74	3	(74, 118, 188)
94	$[X, 24X]$	75	5	(75, 119, 189, 299, 474)
95	$[X, 24X]$	75	6	(75, 119, 189, 299, 474, 751)
96	$[X, 29X]$	75	7	(75, 120, 191, 303, 481, 763, 1210)

5.1. Root branches

Lemma 5.1 (Root branch cover). *For each row of Table 1, the branches*

$$k_1 = 1, \dots, k_{1,\max}$$

are disjoint and exhaust every odd n_1 in the stated window.

Proof. Each odd n_1 determines the unique value $k_1 = \nu_2(n_1 + 1) \geq 1$. The window endpoints give

$$\frac{73}{10}X + 1 < 2^{74}, \quad 15X + 1 < 2^{75}, \quad 24X + 1 < 2^{76}, \quad 29X + 1 < 2^{76}.$$

Thus the displayed maxima 73, 74, 75, 75, 75 are exhaustive. Uniqueness of the valuation makes the branches disjoint. $\square$

The later caps follow from Hercher's growth law. If $n_1 + 1 < 2^B$, then

$$k_i < \delta^{i-1}B < \bar{\delta}^{i-1}B. \quad (19)$$

The exact reduction verifier checks that every encoded cap in Table 1 is at least the largest integer allowed by (19). Some caps are intentionally loose; this only enlarges the search.

5.2. Rotation-correct stage minima

The search remains rotated at the global minimum n_1 . Later lower bounds therefore cannot be obtained by introducing a second suffix-balanced rotation. Instead they follow from the odd-step mass remaining after the preceding valuation caps are subtracted.

Lemma 5.2 (Cap-complement stage bound). *Suppose $K \geq K_0$ and $k_j \leq \kappa_j$ for $1 \leq j < i$. Put $r = m - i + 1$. Then*

$$\log_2(n_i + 1) > \left(K_0 - \sum_{j < i} \kappa_j \right) \frac{\delta - 1}{\delta^r - 1} > \left(K_0 - \sum_{j < i} \kappa_j \right) \frac{\bar{\delta} - 1}{\bar{\delta}^r - 1}. \quad (20)$$

Proof. The remaining r blocks carry

$$\sum_{j=i}^m k_j = K - \sum_{j < i} k_j \geq K_0 - \sum_{j < i} \kappa_j.$$

Apply Lemma 2.3 to the consecutive block beginning at n_i . The second inequality follows because $(x - 1)/(x^r - 1)$ decreases for $x > 1$. $\square$

The exact case generator chooses an integer target N_i only after proving, with rational arithmetic, that

$$\log_2(N_i + 1) < \left(K_0 - \sum_{j < i} \kappa_j \right) \frac{\bar{\delta} - 1}{\bar{\delta}^{m-i+1} - 1}.$$

Thus $n_i \geq N_i$ is necessary. The nontrivial targets are shown in Table 2; unlisted stages use only $n_i \geq X$.

Table 2: Nontrivial certified stage minima used by the engines.	
m	stage minima
92	$n_2 \geq 32274541490524303248485, \quad n_3 \geq 2^{118}$
93	$n_3 \geq 2^{75}, \quad n_4 \geq 7 \cdot 2^{117}$
94	$n_4 \geq 2^{75}, \quad n_5 \geq 2^{119}, \quad n_6 \geq 2^{189}$
95	$n_5 \geq 3 \cdot 2^{74}, \quad n_6 \geq 7 \cdot 2^{117}, \quad n_7 \geq \lceil 93 \cdot 2^{189}/50 \rceil$
96	$n_6 \geq 3 \cdot 2^{74}, \quad n_7 \geq 7 \cdot 2^{117}, \quad n_8 \geq \lceil 93 \cdot 2^{189}/50 \rceil$

All stage bounds in Table 2 are certified consequences of Lemma 5.2; they are not empirical observations from the search.

6. A finite short-prefix condition

The engines use one local rigidity condition. We prove only the finite form required here; the broader surplus-frontier theory is developed separately in [8]. Let

$$T(x) = \frac{3x + 1}{2^{\nu_2(3x+1)}}$$

be the accelerated map on positive odd integers. Consider an accelerated segment

$$x_0, x_1, \dots, x_t, \quad 3x_j + 1 = 2^{b_j} x_{j+1},$$

and put

$$S_t = \sum_{j=0}^{t-1} b_j, \quad F_t = S_t - t\delta, \quad \sigma_t = \sum_{j=0}^{t-1} 2^{F_j}.$$

Lemma 6.1 (Weighted affine identity). *For every realized accelerated prefix,*

$$x_t = 2^{-F_t} \left(x_0 + \frac{\sigma_t}{3} \right). \quad (21)$$

Proof. Iteration of $2^{b_j}x_{j+1} = 3x_j + 1$ gives

$$2^{S_t}x_t = 3^t x_0 + \sum_{j < t} 3^{t-1-j} 2^{S_j}.$$

Since $2^{S_j} = 3^j 2^{F_j}$, the sum equals $3^{t-1} \sigma_t$. Divide by $2^{S_t} = 3^t 2^{F_t}$. $\square$

Theorem 6.2 (Prefix Hugging). *If $x_j \geq x_0$ for $0 \leq j \leq t$, then*

$$F_t \leq t \log_2 \left(1 + \frac{1}{3x_0} \right) < \frac{t}{3x_0 \log 2}. \quad (22)$$

Proof. From (21) and $x_t \geq x_0$,

$$2^{F_t} \leq 1 + \frac{\sigma_t}{3x_0}.$$

Consequently

$$\sigma_{t+1} = \sigma_t + 2^{F_t} \leq 1 + \left(1 + \frac{1}{3x_0} \right) \sigma_t.$$

Starting from $\sigma_0 = 0$, this affine recurrence gives

$$\sigma_t \leq 3x_0 \left[\left(1 + \frac{1}{3x_0} \right)^t - 1 \right].$$

Substitution in the first inequality yields

$$2^{F_t} \leq \left(1 + \frac{1}{3x_0} \right)^t.$$

Taking base-2 logarithms proves the first inequality in (22); the second follows from $\log(1+u) < u$. $\square$

At the end of a block prefix in the shortcut model, put

$$u = \sum k_i, \quad v = \sum \ell_i.$$

The corresponding accelerated prefix has length u and valuation sum $u + v$, so its surplus is $u + v - u\delta$. Because the cycle is rotated at its global least minimum, every accelerated state in the prefix is at least $n_1 \geq X$.

Lemma 6.3 (Finite prefix gate). *Every search prefix satisfies*

$$u + v \leq \lfloor u\delta \rfloor. \quad (23)$$

Proof. The largest possible u in any configured search is the sum of the $m = 96$ caps,

$$75 + 120 + 191 + 303 + 481 + 763 + 1210 = 3143.$$

The common reduction certificate checks, for every integer $1 \leq u \leq 3143$, the strict rational inequality

$$\lceil u\delta \rceil - u\delta > \frac{u}{3X \log 2}. \quad (24)$$

The same certified bracket decides $\lfloor u\delta \rfloor$ throughout this range; the smallest margin occurs at $u = 2966$ and remains positive.

If (23) failed, the integer $u + v$ would satisfy

$$u + v - u\delta \geq \lceil u\delta \rceil - u\delta.$$

On the other hand, Theorem 6.2 gives

$$u + v - u\delta < \frac{u}{3n_1 \log 2} \leq \frac{u}{3X \log 2},$$

contradicting (24). $\square$

This finite lemma is the entire prefix-rigidity input to Theorem 1.1; the much larger certified surplus frontier of [8] is not used here.

7. The affine-ladder search and completeness

Fix a case m , a depth d , and one root value k_1 . Write

$$n_1 = 2^{k_1} a_1 - 1,$$

where a_1 is odd. The initial values of a_1 form one finite odd progression. The search carries all later conditions back to this single variable.

7.1. Recursion invariant

At level i , the parameters $k_1, \dots, k_i$ and $\ell_1, \dots, \ell_{i-1}$ are fixed. The engine maintains integers p, q, s such that

$$a_i = \frac{pa_1 + q}{2^s} \quad (25)$$

for every surviving a_1 , and the live set is represented exactly as

$$\mathcal{L} = \{a_1 \in \mathbb{Z} : a_1 \equiv r \pmod{2^b}, L \leq a_1 \leq U\}. \quad (26)$$

The coefficient p is odd. The initial call has $(p, q, s) = (1, 0, 0)$ and consists exactly of the odd a_1 for which n_1 lies in the selected window.

For a proposed value ℓ at level i , set $k = k_i$ and

$$p' = 3^k p, \quad q' = 3^k q + (2^\ell - 1)2^s. \quad (27)$$

A direct substitution in Lemma 2.1 gives

$$n_{i+1} + 1 = \frac{p'a_1 + q'}{2^{s+\ell}}. \quad (28)$$

In particular, p' remains odd.

7.2. Exact valuation partitions

Lemma 7.1 (ℓ_i -partition). *For a fixed live value a_1 , exactly one branch of the ℓ -loop can survive the condition that n_{i+1} is odd, namely*

$$\ell = \nu_2(3^{k_i} a_i - 1).$$

The computed upper cutoff $\ell_{\max}$ cannot omit an admissible value.

Proof. Let $e = \nu_2(3^{k_i} a_i - 1)$ be the true valuation. By (27),

$$p'a_1 + q' = 2^s (3^k a_i - 1 + 2^\ell).$$

If $\ell < e$, the bracket has valuation exactly ℓ , so the numerator in (28) is not divisible by the additional factor 2 needed to make $n_{i+1} + 1$ even. If $\ell > e$, its valuation is $e < \ell$, and the same divisibility test fails. If $\ell = e$, the bracket is 2^e times the sum of two odd integers and is therefore divisible by 2^{e+1} ; this is the unique branch in which n_{i+1} may be odd.

For completeness of the cutoff, let B_{i+1} be the maximum of the global floor, the stage minimum, and the current lower bound for n_1 . A genuine cycle has $n_{i+1} \geq B_{i+1}$. From

$$n_{i+1} = \frac{3^k(n_i + 1) - 2^k}{2^{k+e}}$$

and the current upper bound for n_i , one obtains an explicit integer T with $2^e \leq T$. The engine sets $\ell_{\max} = \lfloor \log_2 T \rfloor$, so $e \leq \ell_{\max}$. $\square$

Once ℓ is fixed, put $y = p'a_1 + q'$. Then

$$k_{i+1} = \nu_2(y) - (s + \ell). \quad (29)$$

Lemma 7.2 (Hensel partition). *The one-bit lifting walk partitions the surviving progression into disjoint classes on which k_{i+1} has each exact admissible value. The cutoff at the encoded cap cannot omit a genuine cycle.*

Proof. The walk begins by imposing $y \equiv 0 \pmod{2^{s+\ell+1}}$, which is equivalent to $k_{i+1} \geq 1$. Suppose a live class is known modulo 2^E and satisfies $2^E \mid y$. Since p' is odd, replacing a_1 by $a_1 + 2^E$ toggles the E th bit of y . Hence the two lifts modulo 2^{E+1} partition the class into one child with $\nu_2(y) = E$ and one child with $\nu_2(y) \geq E + 1$. Iteration gives disjoint exact-valuation classes and a remaining continuation class.

The hard bound $k_{i+1} \leq \lfloor \log_2(n_{i+1} + 1) \rfloor$ and the certified cap from (19) bound every admissible value. The engine stops only at the smaller of these two valid upper bounds. $\square$

7.3. Interval cuts, deterministic tails, and work units

At each ℓ -branch, (28) is an increasing affine function of a_1 . Intersecting (26) with

$$n_{i+1} \geq N_{i+1}, \quad n_{i+1} \geq n_1$$

therefore gives exact integer interval cuts. The first condition enforces the certified stage bound; the second enforces global minimality. Lemma 6.3 removes only impossible valuation prefixes.

If a live progression contains at most 256 values, the engines simulate each one using exact big-integer arithmetic and compute the true valuations rather than continuing symbolically. At the terminal depth, a hit is recorded exactly when at least one a_1 produces a valid odd n_{d+1} satisfying every imposed necessary condition.

For parallel execution, each root progression is divided by index ranges. The plan verifier checks that the work-unit intervals are pairwise disjoint and that their union is the complete root progression. Subdivision changes neither the represented set nor the mathematical search.

Theorem 7.3 (Fixed-branch completeness). *Fix one case in Table 1 and one root value k_1 . Assume the certified window, caps, stage minima, logarithm bracket, and finite prefix gate. Then the affine-ladder recursion enumerates every a_1 satisfying the reduced necessary conditions exactly once through the*

valuation partitions. A zero-hit result for all work units of that root branch proves that the branch contains no hypothetical cycle.

Proof. Induct on the recursion level. The invariant (25)–(26) holds initially. Lemma 7.1 proves that the ℓ -branches retain each live value in exactly its true spike-valuation branch and that the cutoff is complete. Conditional on that branch, Lemma 7.2 partitions by the exact next odd-run length and proves the cutoff complete. The interval and prefix cuts are necessary conditions, so they remove no cycle. Formula (27) preserves the affine invariant with odd leading coefficient. The deterministic tail computes the same transitions directly, and the terminal test records every remaining admissible value. Finally, the verified index-range partition distributes the root set without gaps or overlaps. Thus the values producing hits are exactly the values surviving the reduced model; if the hit count is zero, the branch is empty. $\square$

Theorem 7.4 (Finite-window exclusion). *For each $m \in \{92, 93, 94, 95, 96\}$, no nontrivial positive m -cycle has its least local minimum in the corresponding window of Table 1.*

Computer-assisted proof. By Lemma 5.1, every least minimum in the window belongs to exactly one of the 372 root branches. The frozen plan partitions those branches into 464 work units. Both exact engines report zero hits for every unit. Apply Theorem 7.3 and the verified work-unit cover. $\square$

8. Certified computation and reproducibility

The authoritative computational artifact is version 1.0.0 of the repository accompanying this paper [9]. The archived release has DOI

10.5281/zenodo.21724745

and its global certificate records the result `NO_M_CYCLE_92_TO_96`. It contains 372 root branches, 464 work units, zero hits, and zero unresolved units.

Each work unit has one canonical result from a C++17/GMP prover and one from a separately implemented Rust/rug verifier. The languages, big-integer layers, progression code, parsers, and executables are distinct. Acceptance requires equality of the complete counter vector, not merely agreement on a zero-hit flag.

The certificate has four layers: (i) exact analytic enclosures and reductions; (ii) the branch and work-unit partitions with coverage checks; (iii) dual-engine

Table 3: Per-case finite-search coverage. Represented counts are exact counts of candidate least minima in the corresponding windows.

m	root branches	work units	represented candidate minima
92	73	73	7,437,727,210,519,691,211,571
93	74	74	16,528,282,690,043,758,247,936
94	75	75	27,153,607,276,500,459,978,752
95	75	129	27,153,607,276,500,459,978,752
96	75	113	33,056,565,380,087,516,495,872
Total	372	464	111,329,789,833,651,885,912,883

Table 4: Aggregate production counters, identical for both engines.

quantity	value
recursive nodes	617,029,186,843
deterministic nodes	15,291,500,002,309
deterministic values simulated	14,748,526,057,831
prefix-gate prunes	1,483,220,130
bound prunes	139,297,496,626
terminal intervals reached	0
maximum integer width	105 bits
hits	0

execution records with source, binary, build, and computation provenance; and (iv) schemas, canonical serialization, adversarial tests, and an immutable manifest.

The theorem artifacts can be authenticated without rerunning the expensive searches by executing

```
python3 -B verify_all.py --profile theorem-artifacts
```

from the release root. Acceptance includes the marker

```
ACCEPT_COMPUTATIONAL_ARTIFACT_SET_M_LE_96
```

A separate profile replays every frozen work unit with the independent Rust engine:

```
python3 -B verify_all.py --profile full-replay --jobs <N>
```

The release archive has SHA-256

38912cce89105d4416680b54099c1efc7373d7313b1cf20cfa700638f03fa31f

and the digest of its top-level SHA256SUMS file is

619f794ae3a570d0457c587a5aad49132d77e941bc2236c02da295d066e1b6d7

Remark 8.1 (Trust boundary). The frozen records are execution certificates, not a formally verified proof trace for every prune. The mathematical trust boundary consists of the published inputs, the human proofs of the denominator reduction and search completeness, and correctness of at least one exact implementation. The second engine, matching counters, mutation tests, and frozen provenance provide independent checks against ordinary implementation and artifact-integrity failures.

9. The second lift and proof of the main theorem

Assume the finite window for a fixed m has been excluded. Since n_1 is the least local minimum, every local minimum is then larger than the upper endpoint $W_m X$. Reapply Proposition 3.3 with $K^* = K_0$ and this stronger common floor.

Table 5: Certified structure of the second complement bound.

m	window floor	selected block length	certified exponent floor
92	$(73/10)X$	91	117 plus one distinguished term
93	$15X$	90	115
94	$24X$	89	179
95	$24X$	90	113
96	$29X$	90	112

For $m = 93, 94, 95, 96$, let s_m and f_m be the block length and exponent floor in Table 5, and put $Y_m = W_m X$. The exact reduction uses

$$\sum_{i=1}^m T(n_i) < \frac{3(m - s_m)}{Y_m} + \frac{3s_m}{2^{f_m} - 1}. \quad (30)$$

Indeed, the $m - s_m$ minima outside the suffix-balanced block exceed Y_m , while each selected minimum exceeds $2^{f_m} - 1$.

For $m = 92$, the first selected term is treated by the sharper distinguished-stage estimate, and the certified bound is

$$\sum_{i=1}^{92} T(n_i) < \frac{30}{73X} + \frac{30000}{17086 \cdot 2^{74}} + \frac{270}{2^{117} - 1}. \quad (31)$$

The first term is the single minimum outside the selected block. For the first block element, the exact cap-complement calculation proves

$$n > \frac{17086}{10000} 2^{74},$$

and the remaining 90 block elements exceed $2^{117} - 1$.

Dividing (30) or (31) by $3K_0\lambda$, where $\lambda < \log 2$ is the certified lower endpoint, gives the five final error bounds stored in the reduction certificates. In every case, the resulting interval for $(K + L)/K$ lies strictly between the adjacent fractions

$$\frac{12261796429850908150604}{7736332199829210068325} < \frac{325919355854421968365}{205632218873398596256}. \quad (32)$$

Their determinant has absolute value 1, and their denominator sum is

$$\begin{aligned} & 7736332199829210068325 + 205632218873398596256 \\ & = 7941964418702608664581 = Q. \end{aligned}$$

Thus Lemma 3.4 gives $K \geq Q$.

For each $m = 92, \dots, 96$, the exact reduction certificate also verifies by integer cross-multiplication that

$$K < 1.4784 m \delta^m < \frac{924}{625} m \left(\frac{317}{200} \right)^m < Q. \quad (33)$$

Proof of Theorem 1.1. Fix $m \in \{92, 93, 94, 95, 96\}$ and suppose that a non-trivial positive m -cycle exists. By Theorem 4.1, $K \geq K_0$. Let n_1 be its least local minimum. If $n_1 \leq W_m X$, the cycle lies in the finite window and is excluded by Theorem 7.4. If $n_1 > W_m X$, every local minimum exceeds $W_m X$, so the second lift and (32) give $K \geq Q$, while (33) gives $K < Q$. Both cases are impossible. $\square$

10. Scope and data availability

The result proved here excludes positive nontrivial m -cycles through $m = 96$. It does not prove the Collatz conjecture, exclude negative cycles, or classify arbitrary nonperiodic trajectories. The cases $m = 92, \dots, 95$ are retained because they belong to the independently completed five-case computation and provide one uniform independently checked artifact. After the later discovery of Wang’s preprint, no priority claim is made for those four exclusion results. The case $m = 96$ is the new cycle-exclusion case. The analytic comparison in (33) remains valid through $m = 98$, but the present artifact contains no certified finite searches for $m = 97$ or $m = 98$, and no theorem about those cases is claimed.

Data and code availability

The frozen source code, analytic certificates, work-unit plans, dual-engine results, verifiers, tests, manifests, and reproduction instructions are archived as *Collatz m-96 Certificates Pipeline*, version 1.0.0 [9], DOI [10.5281/zenodo.21724745](https://doi.org/10.5281/zenodo.21724745). The corresponding tagged GitHub release is available at <https://github.com/MaskoMys/Collatz-m96-Certificates-Pipeline/releases/tag/v1.0.0>.

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The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT in order to assist with language, organization, and presentation. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

- [1] D. Bařina, Improved verification limit for the convergence of the Collatz conjecture, *J. Supercomput.* 81 (2025), Article 810. [doi:10.1007/s11227-025-07337-0](https://doi.org/10.1007/s11227-025-07337-0).
- [2] C. Hercher, There are no Collatz m -cycles with $m \leq 91$, *J. Integer Seq.* 26 (2023), Article 23.3.5.
- [3] C. Hercher, Corrigendum to Article 23.3.5: There are no Collatz m -cycles with $m \leq 91$, *J. Integer Seq.*, June 14, 2026. <https://cs.uwaterloo.ca/journals/JIS/VOL26/Hercher/corrigendum.pdf>.
- [4] J. C. Lagarias, The $3x + 1$ problem and its generalizations, *Amer. Math. Monthly* 92 (1985), 3–23. [doi:10.2307/2322189](https://doi.org/10.2307/2322189).
- [5] J. L. Simons and B. M. M. de Weger, Theoretical and computational bounds for m -cycles of the $3n + 1$ problem, *Acta Arith.* 117 (2005), 51–70. [doi:10.4064/aa117-1-3](https://doi.org/10.4064/aa117-1-3).
- [6] J. L. Simons and B. M. M. de Weger, Theoretical and computational bounds for m -cycles of the $3n + 1$ problem, updated version 1.44, August 31, 2010. <https://deweger.net/papers/%5B35a%5DSidW-3n%2B1-v1.44%5B2010%5D.pdf>.
- [7] X. Wang, Non-existence of Collatz m -cycles for $m \leq 95$, preprint, June 2026. [doi:10.5281/zenodo.21017296](https://doi.org/10.5281/zenodo.21017296).
- [8] N. Mghirbi, Prefix rigidity and surplus frontiers for Collatz cycles, companion preprint, 2026.
- [9] N. Mghirbi, *Collatz m -96 Certificates Pipeline*, version 1.0.0, Zenodo, 2026. [doi:10.5281/zenodo.21724745](https://doi.org/10.5281/zenodo.21724745).